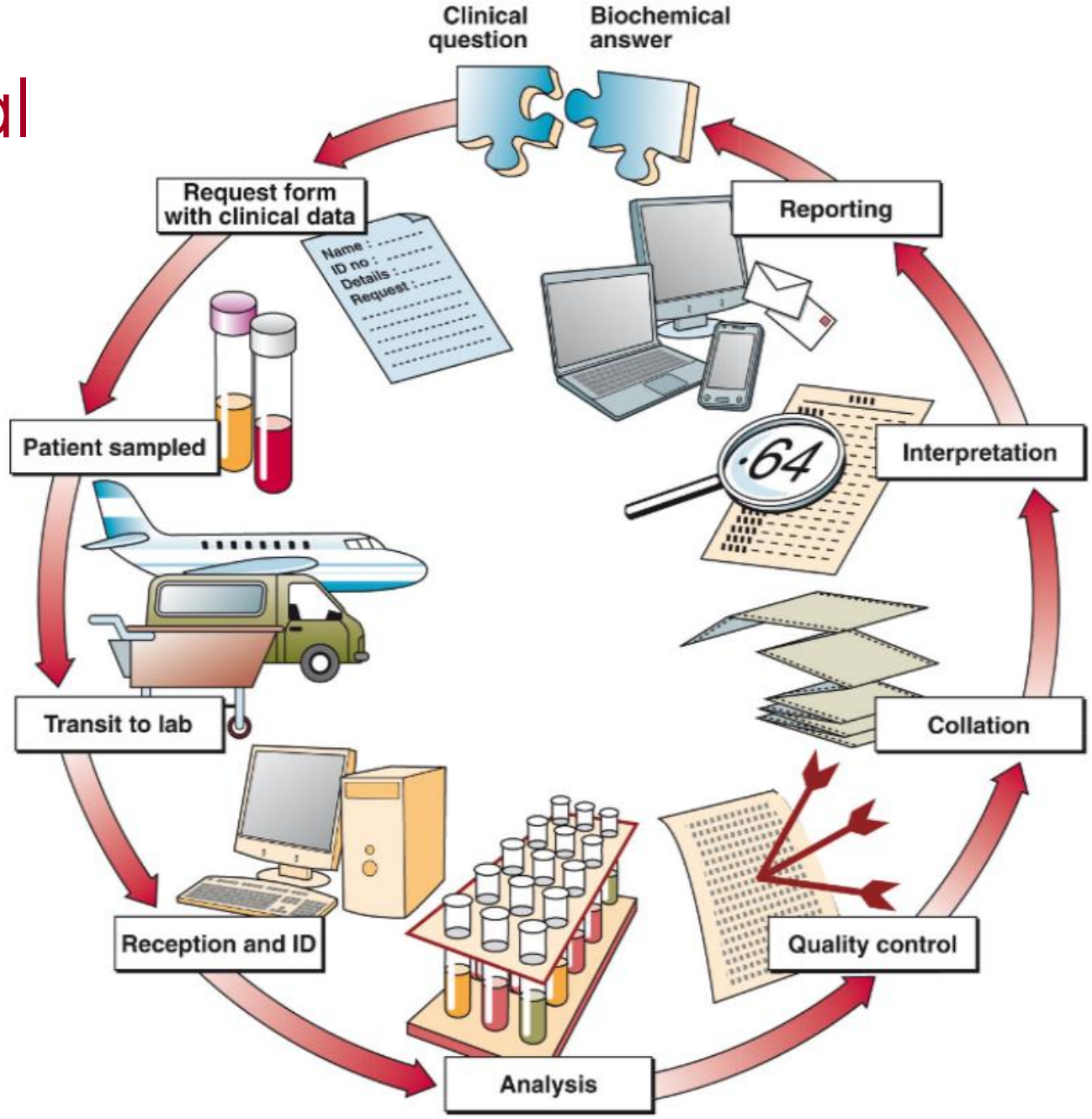
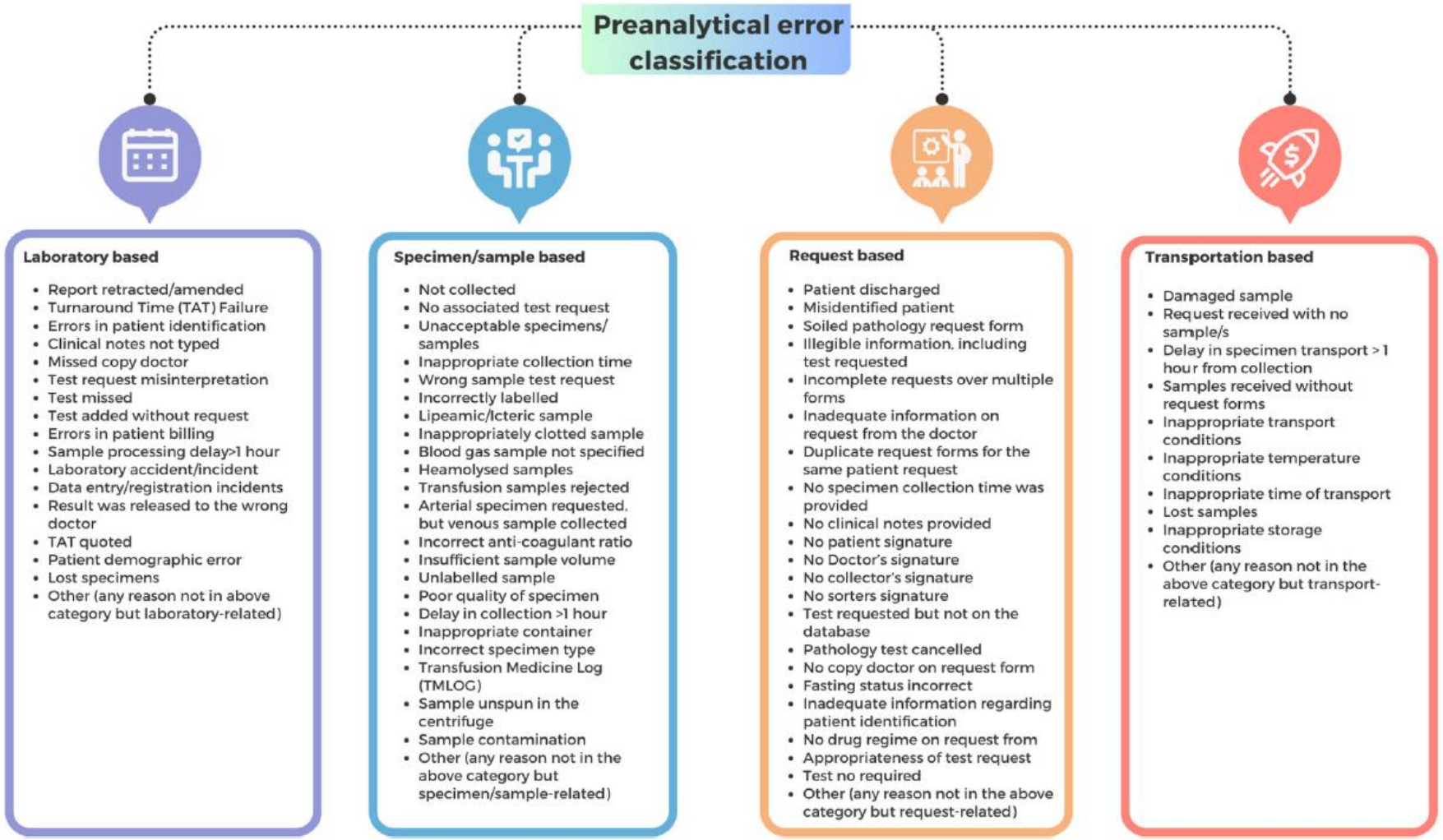
The analytical process





From errors to excellence: the pre-analytical journey to improved quality in diagnostics. A scoping review. George K. John et al. Clin Chem Lab Med 2025; 63(7): 1243–1259. https://doi.org/10.1515/cclm-2024-1277

What are the general recomendations for outpatients preparation prior to blood sampling?

1. Fasting state

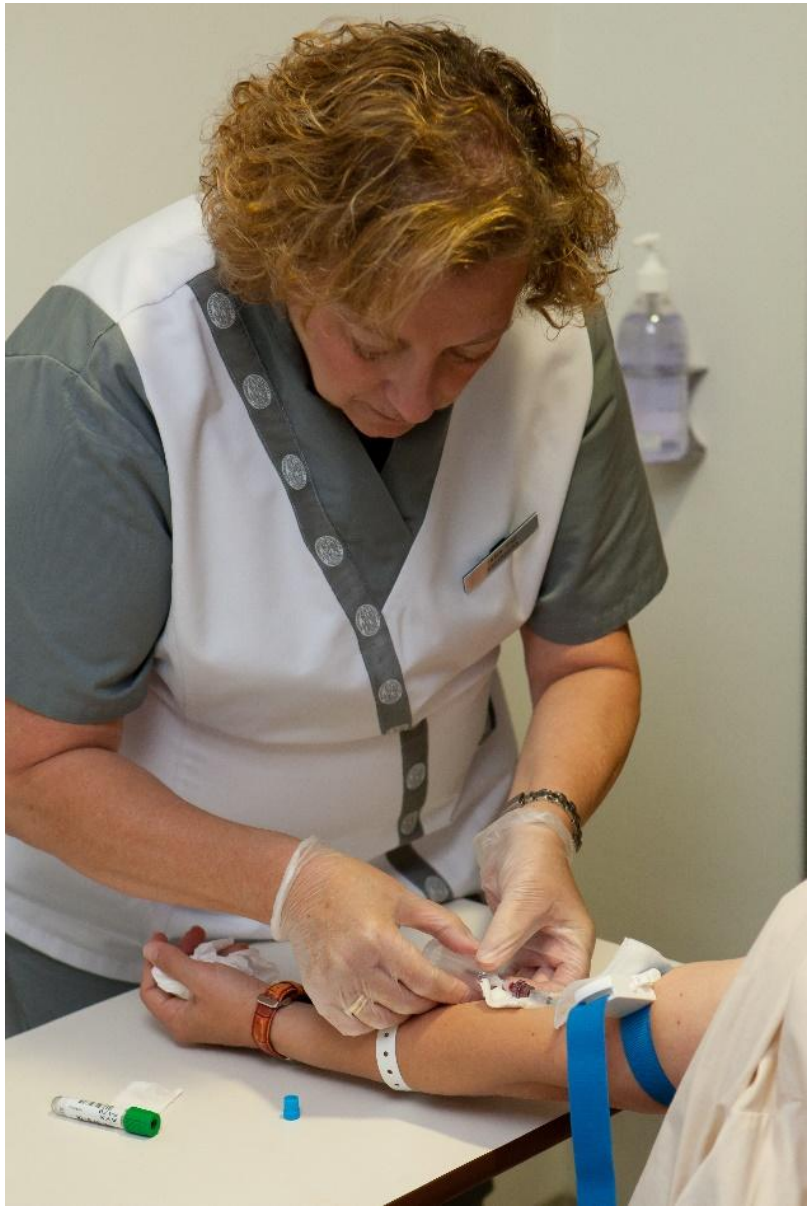
   12 hours after the last meal.

2. Morning collection

   between 7 and 9 am

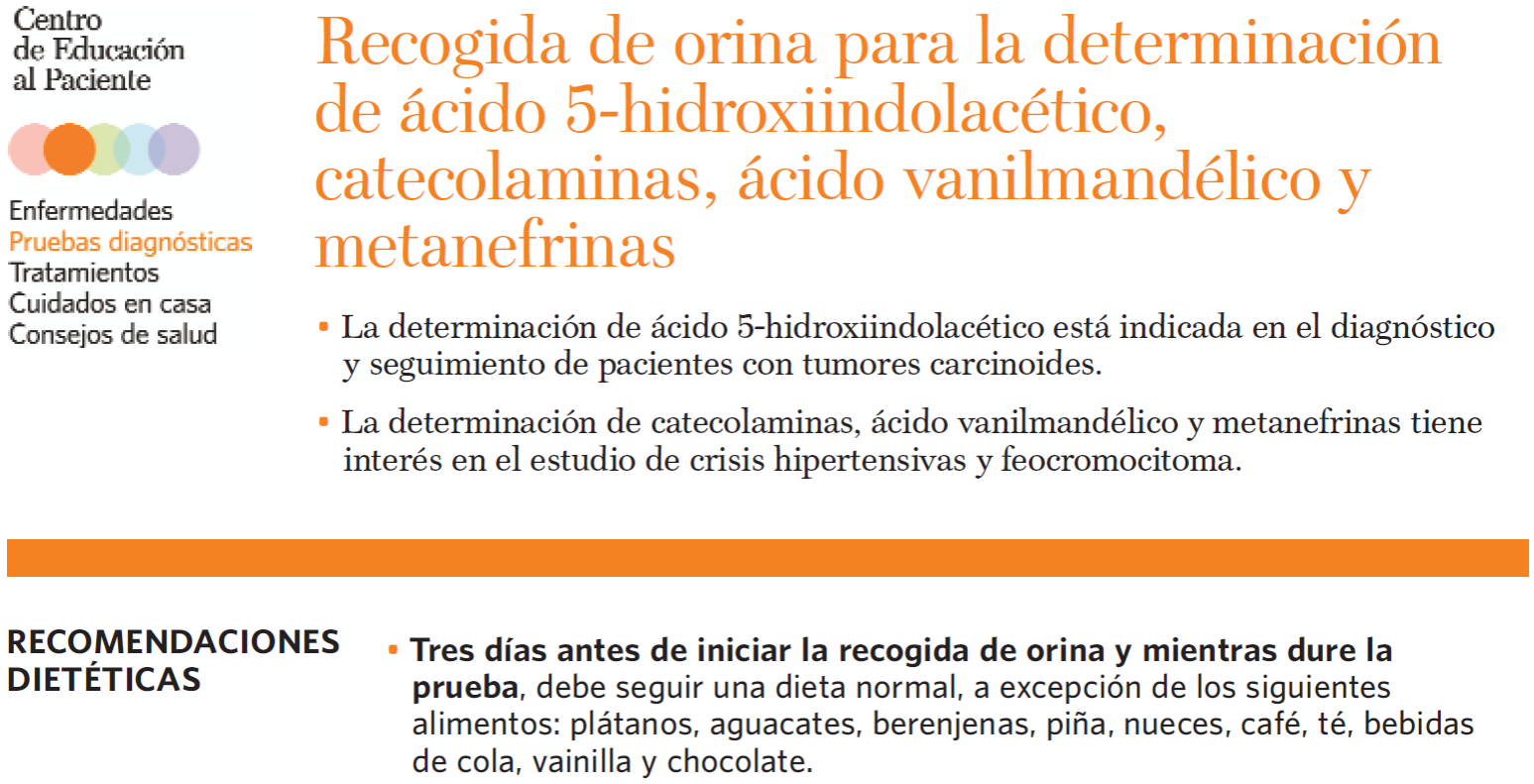
3. No confounding factors

   Exercise
   Shift work
   infections
   Jet lag
   Change of time zome



Recommendations for patient preparation for other samples

Special preparation



Centro de Educación al Paciente

Enfermedades
Pruebas diagnósticas
Tratamientos
Cuidados en casa
Consejos de salud

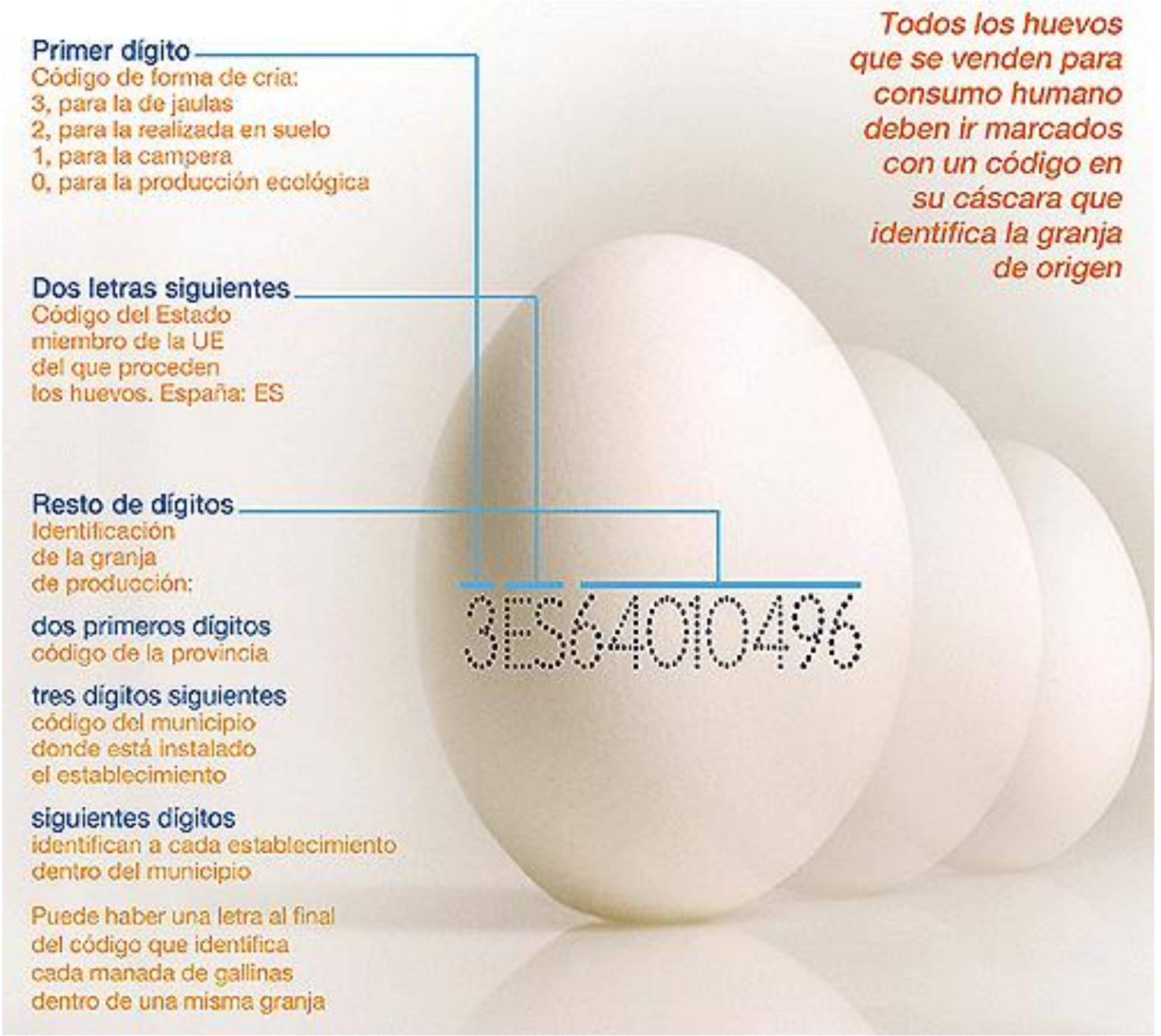
Recogida de orina para la determinación de ácido 5-hidroxiindolacético, catecolaminas, ácido vanilmandélico y metanefrinas

- La determinación de ácido 5-hidroxiindolacético está indicada en el diagnóstico y seguimiento de pacientes con tumores carcinoides.
- La determinación de catecolaminas, ácido vanilmandélico y metanefrinas tiene interés en el estudio de crisis hipertensivas y feocromocitoma.

RECOMENDACIONES DIETÉTICAS

- **Tres días antes de iniciar la recogida de orina y mientras dure la prueba**, debe seguir una dieta normal, a excepción de los siguientes alimentos: plátanos, aguacates, berenjenas, piña, nueces, café, té, bebidas de cola, vainilla y chocolate.

TRACEABILITY



TRACEABILITY

Pacient
Name, last name, medical record
Date of birth/ gender
Hospitalized or outpatient
Doctor, Department

Personnel/Date and time
Request
Extraction
Receptión
Who performed the test
validation

Sample and test
Hemolysis, lipemia, icterus indexes
Comments

Incidences

Results / request
Result and units
Reference values

Analyser

Reagents/Calibrators/Controls
Name
Manufacturer
System
Lot
Expiring date

The barcoded label contains all the information of the process performed to obtain a test result

Tracking and tracing



Blood specimen tubes



TABLA 1-1 Tipos de tubos de muestra que se emplean habitualmente en el laboratorio

	Anticoagulante	Color del tapón	Ejemplos de determinaciones
Suero		Marrón	Enzimas, glucosa, lípidos y otros Iones: Ca^{2+}, Mg^{2+}, Fe^{3+} y Li^{+} Serología: de hepatitis, del VIH, etc. Hormonas, marcadores tumorales y otros
Plasma	EDTA-K_3	Lila	Hormonas Pruebas de urgencias
	Heparina de litio	Verde	Hormonas Electrolitos: Na^{+}, K^{+}, Cl^{-} y HCO_3^{-}
	Fluoruro/oxalato	Gris	Lactato
	Citrato de sodio	Azul	Pruebas de coagulación Velocidad de sedimentación
Sangre	EDTA-K_3	Lila	Glicohemoglobina A1c Hemograma

Anticoagulants:
EDTA: ethylene diamine tetra acetic acid: irreversible calcium quelanting agent - *hemogram*
Lithium heparin: Inhibits thrombin
Citrate: reversible calcium quelanting agent–*coagulation tests*

Preservatives:
NaF: inhibitor of glycolysis – *lactate*
Aprotinin: Proteases inhibitor- rapidly degraded proteins

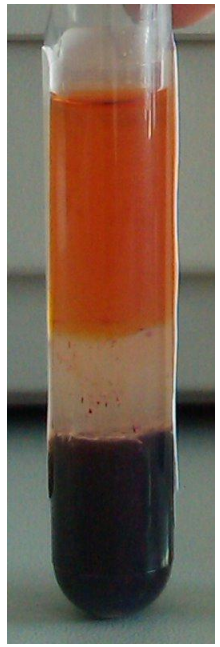
Serum obtention

Non centrifuged specimen

Centrifuged (spun down) specimen



Separating gel



Serum 55-60%

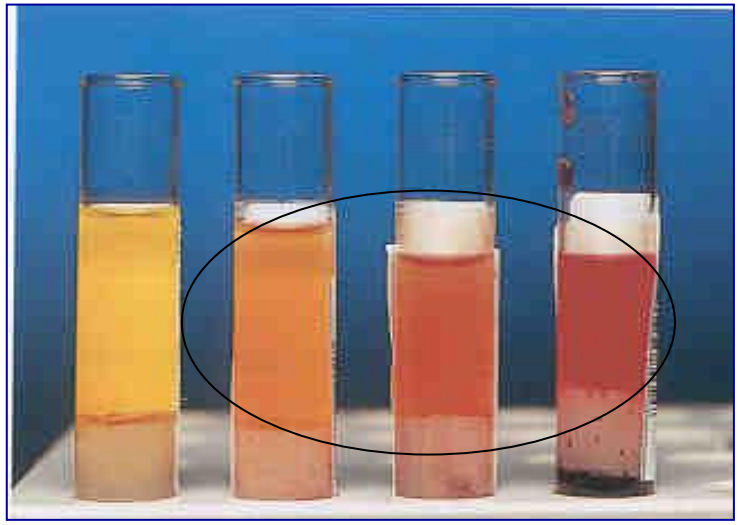
Red cells 40-45%

	SERUM	PLASMA
Use of anticoagulant?		
Let to coagulate for 20 min?		
Requires centrifugation?		
Clotting factors present?		

	SERUM	PLASMA
Use of anticoagulant?	NO	YES
Let to coagulate for 20 min?	YES	NO
Requires centrifugation?	YES	YES
Clotting factors present?	NO	YES

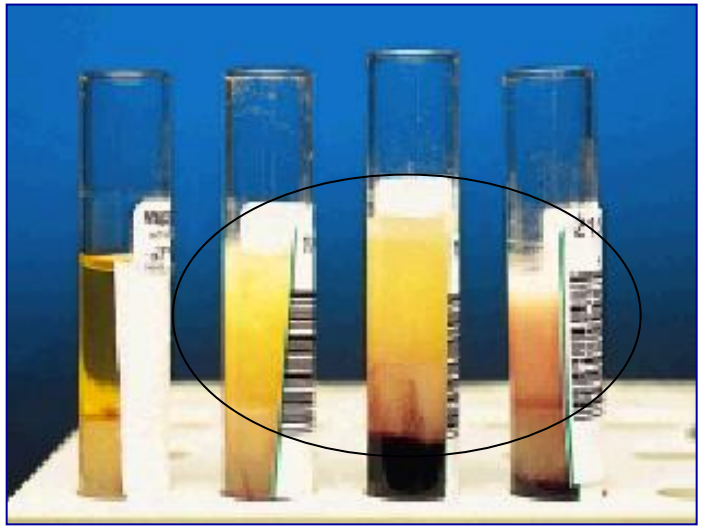
S.N.	Characteristics	Serum	Plasma
1.	Definition	Serum is the liquid part of the blood after the coagulation.	Plasma is a clear and yellowish fluid part of the blood.
2.	Composition	Serum is the water fluid from blood without the clotting factors.	Plasma is the blood fluid that contains blood clotting agents.
3.	Volume	Volume of serum is less in comparison to plasma.	Plasma is a clear yellow liquid which is 55% of total volume of blood.
4.	Isolation	Serum is acquired from the process of spinning after clotting.	Plasma is acquired from the process of spinning before clotting.
5.	Isolation Procedure	Serum is more difficult and time consuming to separate.	Plasma is easier and less time consuming to separate as compared with serum.
6.	Use of Anticoagulants	Serum does not need anticoagulants for separation.	Anticoagulants are necessary to separate plasma.
7.	Composition	Serum contains proteins, electrolytes, antibodies, antigens and hormones.	Plasma is considered as the medium of blood in which RBCs (Red Blood Cells), WBC (White Blood Cells) and other components of blood are suspended.

Analytical interferences due to the physical characteristics of the samples



hemolysis

K, LDH



lipemia

Na



fibrin

False negative

Interferent: Factor different to the analyte to be measured that changes the measured results

No visual inspection, we employ indexes

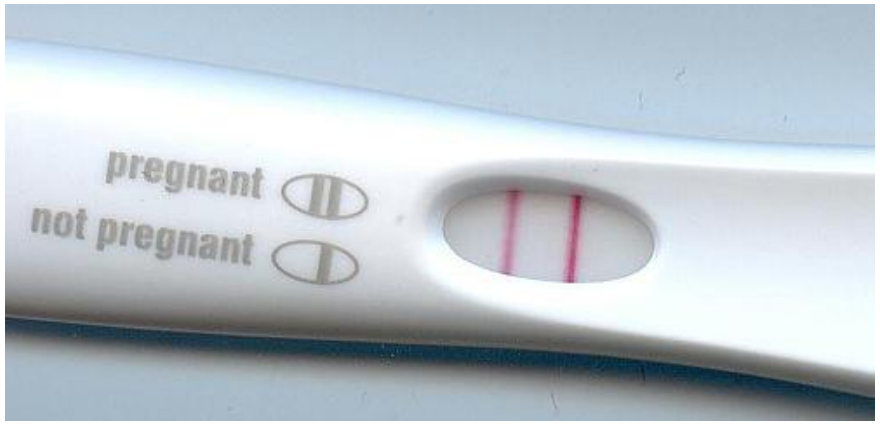
Analytical sensitivity and specificity

Analytical sensitivity

how little of the analyte the method can detect.

Analytical specificity

how good the assay is at discriminating between the requested analyte and potentially interfering substances



Limit of detection



Definition
Examples: HCG for pregnancy vs tumour
Pregnancy test: pharmacy vs laboratory

Analytical interference

Effect of a substance present in the analytical system that causes a deviation in the measured value compared with the true value

1. Endogen
    - Hemolysis, lipemia and icteria
    - Proteins
        - Antibodies
        - Fibrin
2. Exogen
    - Drugs
    - Carry-over

When to suspect an analytical interference?

1. Disagreement between the biochemical pattern and the clinical data

2. Disagreement between two variables physiologically dependent

3. Unlikely/unbelievable results

4. Important changes in the biochemical pattern without variations in the treatment nor in the clinical profile

Some interferences

1. Clasic: hemoglobin bilirrubin, lipemia.
2. Hydroxiurea and urea
3. Inmunotherapy with the immunofixation
4. Anti-insulin antibodies and insulin

Have you ever performed a laboratory test?



Point of Care Testing (POCT)



Glucose in capillary blood

Point of Care Testing



Advantages and limitations of POC

FASTER RESULTS

SIMPLER

PORTABILITY

PATIENTS OUTCOMES



INCREASED COST

PERSONNEL: WORK LOAD, LACK OF EXPERTISE

QC

PREANALYTICAL ERRORS IN POCT

Incorrect Patient Identification
Incorrect assignment of results; verify identity using at least two identifiers.

Incorrect Sample Collection Technique
Excessive pressure, inadequate volume, air bubbles, or non-validated samples.

Use of Materials Under Inappropriate Conditions
Expired materials, non-balanced heparin syringes, or improper storage conditions.

Insufficient Staff Training
Errors in collection, handling, and standardized preanalytical procedures.

Delays or Inadequate Conditions Before Analysis
Delays, temperature exposure, and inadequate sample mixing.

Sample Contamination
Incorrect containers, IV-line dilution, or insufficient flushing.

Hemolysis
Improper collection, transport, handling, or delayed processing.

Presence of Clots
Inadequately anticoagulated or poorly mixed samples causing invalid results.

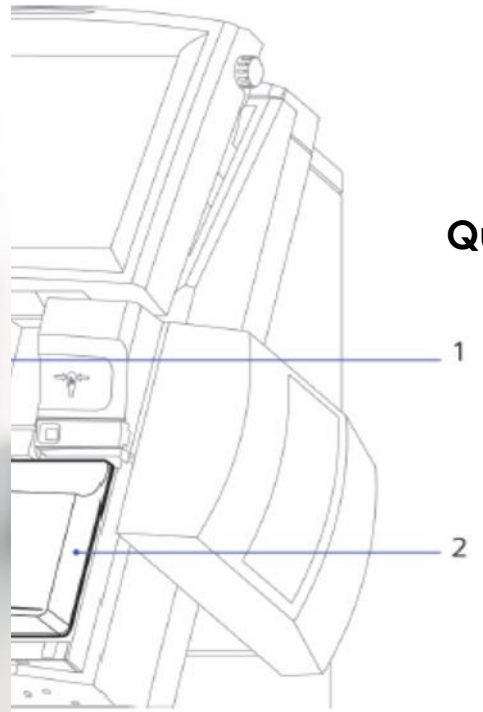
☑ CONCLUSION: Proper patient identification, correct sample collection and handling, appropriate materials, and adequate staff training are essential to prevent most POCT preanalytical errors.

Point of Care Testing

- Urianalysis
- Urine drug tests
- Blood gas analysis
- Capillary glucose
- “Fast” tests for COVID

Blood gas analysis



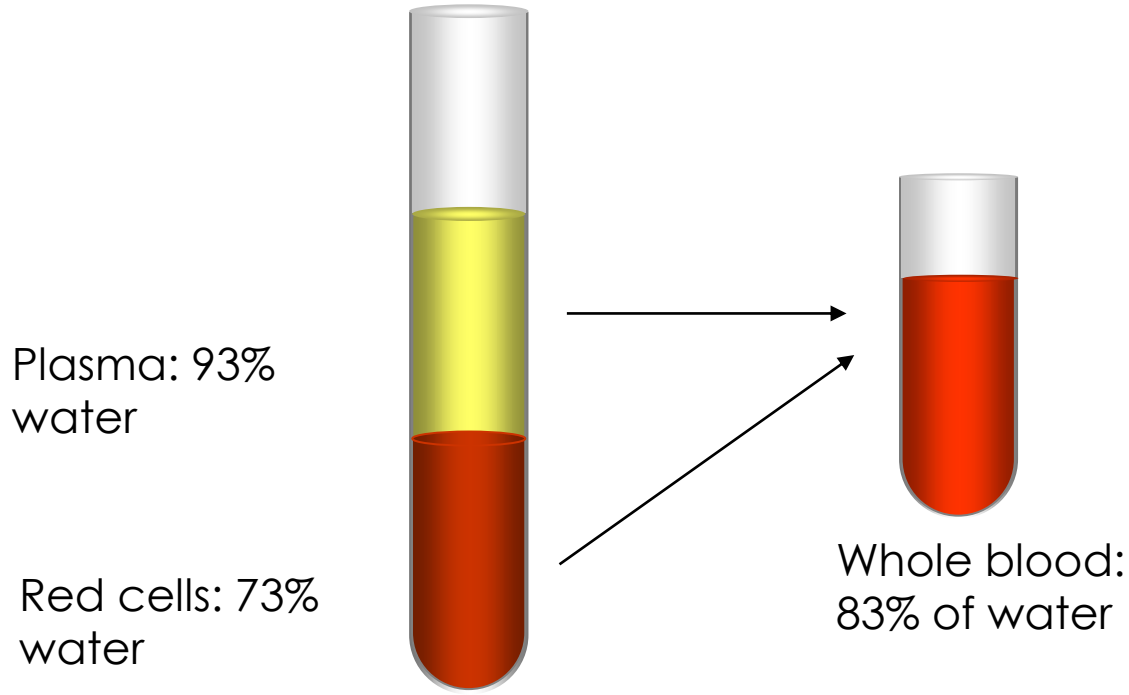


Quality Control Cartridge

Wash/waste cartridge

Whole blood vs plasma

Difference in analyte concentration in whole blood and plasma
Differences are higher when the hematocrit is higher
Different reference values



Water soluble compounds such as glucose or electrolytes are 10% more concentrated in plasma than in whole blood.